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*I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded*

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## 1. Introduction

### Introduction of the business and its forte

#### Business overview

Ms. Mary plans to launch the “E-classroom platform” which is an online application for collage which manages students, teachers, and programs. As a database designer my work is to help her achieve a digital studying environment by creating a strong database system. The platform motto is to enhance the learning experience for students and provide necessary tools for teachers to facilitate effective teaching. By using technology, the EClassroom Platform will run educational processes smoothly, making them more efficient and accessible.

#### Forte

The forte of the E-Classroom Platform

1. **Personalized learning:** The platform allows students to access their modules, assessments, and resources in a systematic manner. Focuses on connecting student's interests with what they learn. It makes the learning process more meaningful for students.
2. **Flexible Curriculum Management:** This provides students a wide range of learning opportunities aligned to their interests and career goal.
3. **Performance evaluation:** The platform provides features for teachers to create and manage assessments linked to specific modules.
4. **Resource Management:** The platform ensure teachers can access all the resources required for teaching which supports students learning.
5. **Communication and Engagement:** The platform is designed that allow teachers and students to communicate.



## Description of Current Business Activities and Operations

In the era of online education Ms. Mary is facing challenges of traditional educational system. The E-Classroom platform is designed to fulfil the modern and current needs of students, providing a complete solution that upgrades the educational experience for both students and teachers.

### Activities and operations

1. **Students Management System:** The platform will have a database for registration of students and evaluate academic progress.
2. **Arranged Program and Module:** The college provides different programs each consisting of various modules. The platform will help students to understand which modules are required for their chosen program and what they need to complete to achieve their academic goals.
3. **Teacher Assignments and Management:** The platform will have a feature for the teacher to upload the assignments linked to specific modules which ensure that teachers can provide the necessary support and guidance to their students.
4. **Assessment Management:** The E-Classroom Platform will let teachers to supervise assessment linked to specific modules. This feature will include essential details such as assessment id, title, deadline, and weightage, enabling students to understand the expectations and requirements for each assessment.
5. **Result Accessibility:** Students will have access to their assessment results through the platform according to the modules. It will help students to evaluate their progress and recognize field for improvement.
6. **Resources for teachers:** The platform will provide teachers with educational resources that will support the learning objectives of each module.
7. **Communication and engagement Tools:** Teachers can post announcement, updates, and important notices related to their modules with will ensure students

are always up to date about the deadlines, events and opportunities for engagement.

## **Business Rules**

1. One program can include many modules. One module is included in many programs.
2. One module links many assessments. One assessment is linked to one module.
3. One assessment can be taken many students. One student can take many assessments.
4. One student can obtain many results. One result is obtained by one student.
5. One module can be taught by many teachers. A teacher can teach many modules.
6. One teacher can access many resources. One resource can be accessed by many teachers.

## **2.Initial ERD**

### **Identification of Entities and Attributes**

#### **Entity**

An entity is a conceptual or logical representation of a real-world object that stores data in a database. (Oyairo, 2024)Any real-world object or concept for which we maintain data can be classified as an entity. These entities are represented in database and must be identified with specific context.

#### **Attributes**

An attribute is a characteristic or property of an entity. It describes the entity in more detail. We can visually represent an attribute as an oval shape connected to the entity it belongs to in an ER diagram. (S, 2024)

*Table 1:Entities and Attributes*

| <b>Entities</b> | <b>Attributes</b>                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------|
| Programs        | Program_id, Program_name,<br>Degree_level                                                                |
| Modules         | Module_id, Program_id, Module_name,<br>Enrollment_capacity                                               |
| Assessments     | Assessment_id, Assessment_title,<br>Deadline, Weightage, Module_id                                       |
| Students        | Student_id, Student_name, Date_of_birth,<br>Gender, Program_id, Enrollment_date,<br>Email, Assessment_id |
| Result          | Grade, Date_obtained,<br>Student_id, Assessment_id, Program_id,<br>Module_id                             |
| Teachers        | Teacher_id, Name, Email, Phone_number,<br>Hire_date, Module_id                                           |
| Resources       | Resource_id, Title, Type, Duration,<br>Module_id, Teacher_id                                             |

### Identification and representation of the Primary keys

A primary key is a unique identifier assigned to each record in a database table. It ensures that every record is distinct, preventing duplication and maintaining data integrity. By using primary keys, we can efficiently locate, update, and delete specific records. Additionally, they help organize the database structure, making it easier to navigate and query data. (Melanie, 2024)

*Table 2: Identification of Primary key and foreign key*

| Entities    | Primary key   | Foreign key                              |
|-------------|---------------|------------------------------------------|
| Programs    | Program_id    | College_id                               |
| Modules     | Module_id     | Program_id                               |
| Assessments | Assessment_id | Module_id                                |
| Students    | Student_id    | Assessment_id<br>Module_id               |
| Result      |               | Assessment_id<br>Module_id<br>Student_id |
| Teachers    | Teacher_id    | Module_id                                |
| Resources   | Resource_id   | Module_id<br>Teacher_id                  |

## Entity Relationship Diagram

An Entity Relationship Diagram is a visual representation of how different entities are connected in database. It's a fundamental tool in database design. In today's business environment, ER Diagrams are used to capture user requirements and then provide a blueprint for databases administrators to create the actual database. (S, 2024)

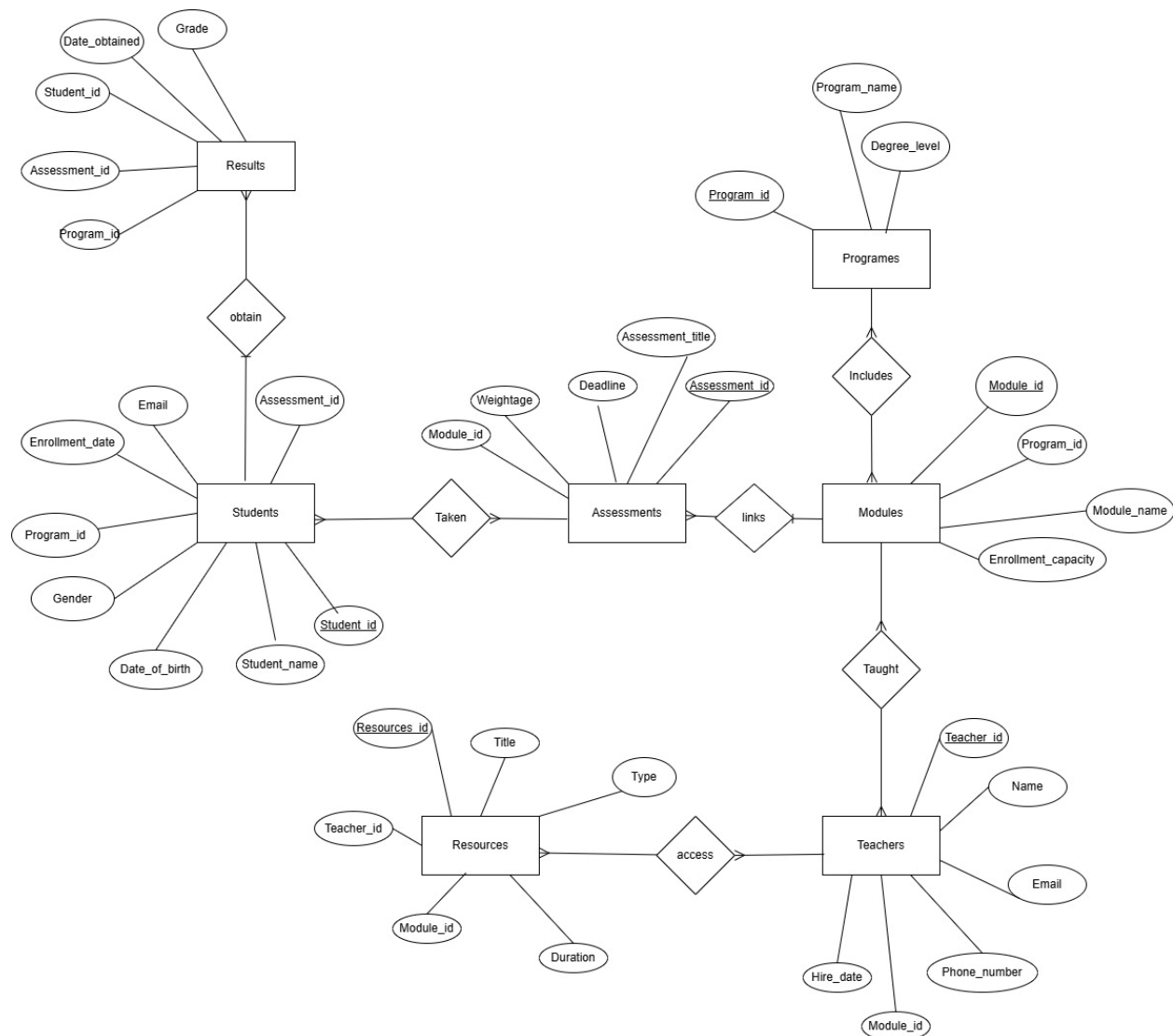


Figure 1:Entities relationships Diagram

### 3.Normalization

Normalization is the method of structuring data within a database to enhance organization. This involves designing tables and defining relationships between them according to specific principles aimed at safeguarding the data and improving database flexibility by reducing redundancy and inconsistent dependencies. (Qiu1987, et al., 2024)

#### Normalization from UNF to 3NF

##### Un-Normalization From (UNF)

Unnormalized Form (UNF), also referred to as an unnormalized relation or a non-first normal form (NF2) relation, is a basic method of representing data in a data model. It is the simplest structure for organizing tables in a database and does not adhere to any normalization principles. UNF is marked by redundant data and frequently includes complex data structures within a single attribute. (Garge, 2024)

- Programs (Program\_id, Program\_name, Degree\_level, Student\_id, Student\_name, Student\_DOB, Gender, Student\_enrollment\_date, Student\_email, {Program\_id, Module\_id, Module\_name, Module\_enrollment\_capacity, Teacher\_id, Teacher\_name, Teacher\_email, Phone\_number, Hire\_date, Assessment\_id, Assessment\_title, Deadline, Weightage, Resources\_id, Resources\_title, Resources\_type, Resources\_duration, Grade, Date\_obtained ,Status})

##### 1<sup>st</sup> Normal Form (1NF)

- Programs (Program\_id, Program\_name, Degree\_level, Student\_id, Student\_name, Student\_DOB, Gender, Student\_enrollment\_date, Student\_email)
- Module\_enrollment (Program\_id, Module\_id, Module\_name, Module\_enrollment\_capacity, Teacher\_id, Teacher\_name, Teacher\_email,

Phone\_number, Hire\_date, Assessment\_id, Assessment\_title, Deadline,  
 Weightage, Resources\_id, Resources\_title, Resources\_type,  
 Resources\_duration, Grade, Date\_obtained)

## 2<sup>nd</sup> Normal Form (2NF)

- Programs (Program\_id, Program\_name, Degree\_level, Student\_id, Student\_name, Student\_DOB, Gender, Student\_enrolment\_date, Student\_email, Grade)
- For Module\_enrollment (Program\_id, Module\_id, Program\_id, Module\_id, Module\_name, Module\_enrollment\_capacity, Teacher\_id, Teacher\_name, Teacher\_email, Phone\_number, Hire\_date, Assessment\_id, Assessment\_title, Deadline, Weightage, Resources\_id, Resources\_title, Resources\_type, Resources\_duration, Grade, Date\_obtained)
- Program\_id->
- Module\_id-> Module\_details  
 Modules (Program\_id, Module\_id, Module\_name, Module\_enrollment\_capacity, Assessment\_id, Assessment\_title, Deadline, Weightage)
- Teacher\_id->Teacher\_details  
 Teachers (Module\_id, Teacher\_id, Teacher\_name, Teacher\_email, Phone\_number, Hire\_date)
- Program\_id, Module\_id->  
 Program\_Modules (Program\_id, Module\_id)
- Teacher\_id, Module\_id->

Teacher\_Module (Module\_id, Teacher\_id, Resources\_id, Resources\_title, Resources\_type, Resources\_duration)

- Result (Student\_id, Assessment\_id, Program\_id, Module\_id, Grade, Date\_obtained)

### 3<sup>rd</sup> Normal Form

- Programs (Program\_id, Program\_name, Degree\_level)
- Students (Student\_id, Student\_name, Student\_DOB, Gender, Student\_enrolment\_date, Student\_email, Assessment\_id)
- Program\_Students (Program\_id, Student\_id)
- Modules (Module\_id, Module\_name, Module\_enrollment\_capacity)
- Program\_Modules (Program\_id, Module\_id)
- Teachers (Teacher\_id, Teacher\_name, Teacher\_email, Phone\_number, Hire\_date)
- Teacher\_Modules (Module\_id, Teacher\_id)



- Resources (Resources\_id, Resources\_title, Resources\_type, Resources\_duration, Teacher\_id, Module\_id)
- Teacher\_Resources (Teacher\_id, Resources\_id)
- Assessments (Assessment\_id, Assessment\_title, Deadline, Weightage, Module\_id)
- Student\_assessments (Assessment\_id, Student\_id)
- Result (Student\_id, Assessment\_id, Program\_id, Module\_id, Grade, Date\_obtained,)

**Therefore, the tables that were obtained after the normalization are:**

- Programs
- Students
- Program\_Students
- Modules
- Program\_Modules
- Teachers
- Teacher\_Modules
- Resources
- Teacher\_Resources
- Assessments

## ➤ Result

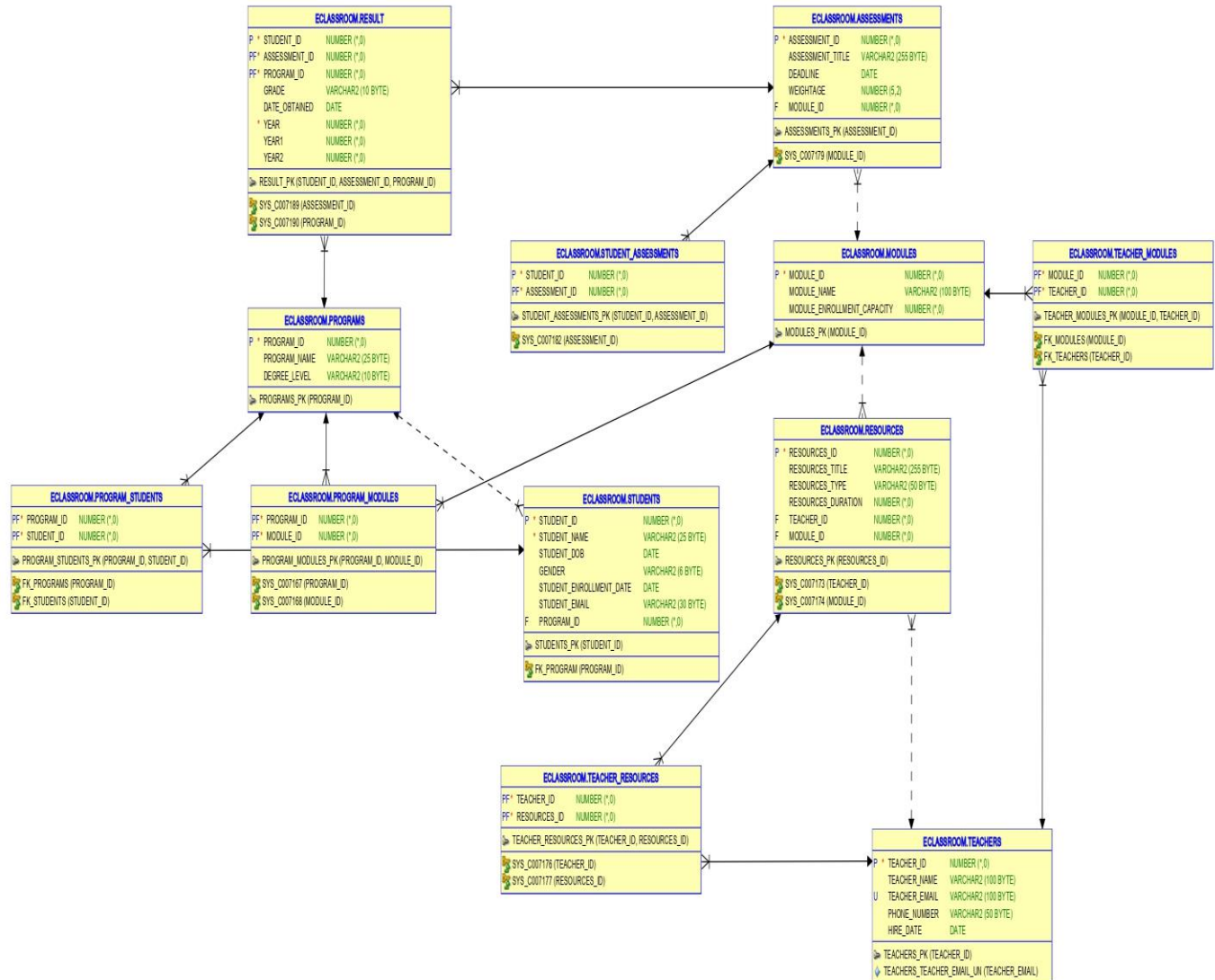
**4.Final ERD**

Figure 2:Final ERD

## 5.Database Implementation

The first step in building our database involves creating a user. Using Oracle SQL and SQL Developer, I developed a database named “Eclassroom”, designed as an online platform. This database is structured to effectively manage programs, student information, modules, teachers, and assessments. It has been configured to comply with business rules and support smooth operations within the Oracle SQL Developer environment.

```
SQL> CREATE USER Eclassroom IDENTIFIED BY tilasmi;  
User created.  
SQL> GRANT RESOURCE, CONNECT TO Eclassroom;  
Grant succeeded.  
SQL>
```

*Figure 3: User created*

```
SQL> connect  
Enter user-name: Eclassroom  
Enter password:  
Connected.  
SQL> |
```

*Figure 4: User connected*

After creating the user, we proceed to design the database by creating the necessary tables. The list of tables to be created is outlined below:

- Programs
- Students
- Program\_students
- Modules
- Program\_Modules
- Teachers
- Teacher\_Modules
- Resources
- Teacher\_Resources
- Assessments
- Result

### Table Creation

The tables created in the database, as detailed, as detailed earlier following the normalization process, are as follows:

#### Creating Programs Table

```
SQL> CREATE TABLE Programs (  
2      Program_id INT PRIMARY KEY,          -- Unique identifier for each  
program  
3      Program_name VARCHAR(100) NOT NULL,  -- Name of the program  
4      Degree_level VARCHAR(50) NOT NULL    -- Level of the degree (e.g.,  
Bachelor, Master)  
5 );  
  
Table created.
```

*Figure 5: Creating Program Table*

### Creating Students Table

```
SQL> CREATE TABLE Students (  
2     StudentID INT PRIMARY KEY,  
3     FirstName VARCHAR(50) NOT NULL,  
4     LastName VARCHAR(50) NOT NULL,  
5     BirthDate DATE,  
6     Email VARCHAR(100) UNIQUE,  
7     EnrollmentDate DATE DEFAULT CURRENT_DATE  
8 );
```

Table created.

Figure 6:Creating Students Table

```
SQL> ALTER TABLE Students  
2  ADD CONSTRAINT fk_program  
3  FOREIGN KEY (program_id)  
4  REFERENCES Programs(program_id);
```

Table altered.

Figure 7: Altered Students table

### Creating Program\_students Table

```
SQL> CREATE TABLE Program_students (  
2     Program_id INT,  
3     College_id INT,  
4     PRIMARY KEY (Program_id, College_id)  
5 );
```

Table created.

Figure 8:Creating Program\_students Table

```
SQL> ALTER TABLE Program_students
  2  ADD CONSTRAINT fk_programs
  3  FOREIGN KEY (Program_id)
  4  REFERENCES Programs(Program_id);

Table altered.

SQL> ALTER TABLE Program_students
  2  ADD CONSTRAINT fk_students
  3  FOREIGN KEY (student_id)
  4  REFERENCES Students(student_id);

Table altered.

SQL> commit;

Commit complete.

SQL>
```

Figure 9: Altering Program\_students Table

### Creating Modules Table

```
SQL> CREATE TABLE Modules (
  2      Module_id INT PRIMARY KEY,
  3      Module_name VARCHAR(100),
  4      Module_enrollment_capacity INT
  5  );

Table created.
```

Figure 10: Creating Modules Table

```
SQL> ALTER TABLE Modules
  2  ADD Date_Announcements DATE;

Table altered.

SQL> |
```

Figure 11: Altered Modules table

**Creating Program\_Moduels Table**

```
SQL> CREATE TABLE Program_Modules (  
2     Program_id INT,  
3     Module_id INT,  
4     PRIMARY KEY (Program_id, Module_id),  
5     FOREIGN KEY (Program_id) REFERENCES Programs(Program_id),  
6     FOREIGN KEY (Module_id) REFERENCES Modules(Module_id)  
7 );  
  
Table created.
```

*Figure 12:Creating Program\_Moduels Table***Creating Teachers Table**

```
SQL> CREATE TABLE Teachers (  
2     Teacher_id INT PRIMARY KEY,  
3     Teacher_name VARCHAR(100),  
4     Teacher_email VARCHAR(100) UNIQUE,  
5     Department VARCHAR(50),  
6     Hire_date DATE  
7 );  
  
Table created.
```

*Figure 13:Creating Teachers Table***Creating Teacher\_Modules Table**

```
SQL> CREATE TABLE Teacher_Modules (  
2     Module_id INT,  
3     Teacher_id INT,  
4     PRIMARY KEY (Module_id, Teacher_id)  
5 );  
  
Table created.
```

*Figure 14:Creating Teacher\_Modules Table*

```
SQL> ALTER TABLE Teacher_modules
 2 ADD CONSTRAINT fk_modules
 3 FOREIGN KEY (Module_id)
 4 REFERENCES Modules(Module_id);

Table altered.

SQL> ALTER TABLE Teacher_Modules ADD CONSTRAINT fk_teachers FOREIGN KEY (Teacher_id)
 2 REFERENCES Teachers(Teacher_id);

Table altered.

SQL>
```

Figure 15: Altering Teacher\_Modules

### Creating Resources Table

```
SQL> CREATE TABLE Resources (
 2     Resources_id INT PRIMARY KEY,
 3     Resources_title VARCHAR(255),
 4     Resources_type VARCHAR(50),
 5     Resources_duration INT,
 6     Teacher_id INT,
 7     Module_id INT,
 8     FOREIGN KEY (Teacher_id) REFERENCES Teachers(Teacher_id),
 9     FOREIGN KEY (Module_id) REFERENCES Modules(Module_id)
10 );

Table created.
```

Figure 16: Creating Resources Table

### Creating Teacher\_Resources Table

```
SQL> CREATE TABLE Teacher_Resources (
 2     Teacher_id INT,
 3     Resources_id INT,
 4     PRIMARY KEY (Teacher_id, Resources_id),
 5     FOREIGN KEY (Teacher_id) REFERENCES Teachers(Teacher_id),
 6     FOREIGN KEY (Resources_id) REFERENCES Resources(Resources_id)
 7 );

Table created.
```

Figure 17: Creating Teacher\_Resources Table



### Creating Assessments Table

```
SQL> CREATE TABLE Assessments (  
2     Assessment_id INT PRIMARY KEY,  
3     Assessment_title VARCHAR(255),  
4     Deadline DATE,  
5     Weightage DECIMAL(5,2),  
6     Module_id INT,  
7     FOREIGN KEY (Module_id) REFERENCES Modules(Module_id)  
8 );
```

Table created.

*Figure 18:Creating Assessments Table*

### Creating Student\_assessments Table

```
SQL> CREATE TABLE Student_Assessments (  
2     Student_id INT,  
3     Assessment_id INT,  
4     PRIMARY KEY (Student_id, Assessment_id),  
5     FOREIGN KEY (Student_id) REFERENCES Students(Student_id),  
6     FOREIGN KEY (Assessment_id) REFERENCES Assessments(Assessment_id)  
7 );
```

Table created.

*Figure 19:Creating Student\_assessments Table*

### Creating Result Table

```
SQL> CREATE TABLE RESULT (  
2     STUDENT_ID NUMBER,  
3     ASSESSMENT_ID NUMBER,  
4     PROGRAM_ID NUMBER,  
5     GRADE NUMBER,  
6     PRIMARY KEY (STUDENT_ID, ASSESSMENT_ID)  
7 );
```

Table created.

*Figure 20:Creating Result Table*

```
SQL> ALTER TABLE RESULT
  2  ADD (MODULE_ID NUMBER);

Table altered.
```

Figure 21: Altering result table

### Creating Announcements Table

```
SQL> CREATE TABLE Announcements (
  2  Announcement_ID INT PRIMARY KEY,
  3  Module_ID INT,
  4  Announcement_Date DATE,
  5  Announcement_Text VARCHAR(255),
  6  FOREIGN KEY (Module_ID) REFERENCES Modules(Module_ID)
  7 );

Table created.
```

Figure 22: Creating Announcements table

## 6.Insert Statement

The SQL insert statement plays a crucial role in adding one or more records to a table in a relational database. It enables the efficient insertion of new data while ensuring the database remains complete and maintains its integrity.

### Inserting into Programs Table

```
SQL> INSERT INTO Programs (Program_id, Program_name, Degree_level) VALUES
  2  (1, 'Computer Science', 'Bachelor'),
  3  (5, 'Biotechnology', 'Master'),
  4  (3, 'Electrical Engineering', 'Master'),
  5  (2, 'Mechanical Engineering', 'Bachelor'),
  6  (4, 'Civil Engineering', 'Bachelor'),
  7  (6, 'Chemistry', 'PhD'),
  8  (7, 'Physics', 'Bachelor');
```

Figure 23: Inserting into Programs Table

**Inserting into Students Table**

```
SQL> INSERT INTO Students (Student_id, Student_name, Student_DOB, Gender, Student_enrolment_date, Student_email) VALUES  
2 (100000001, 'Aasal Paudel', '2000-05-15', 'male', '2020-09-01', 'aasal.paudel@email.com'),  
3 (100000002, 'Ramesh Khadka', '2003-11-22', 'Male', '2020-09-05', 'ramesh.khadka@email.com'),  
4 (100000003, 'Kritika Khadka', '2004-09-02', 'female', '2021-09-10', 'kritika.khadka@email.com'),  
5 (100000004, 'Biraj Bhatta', '2000-03-06', 'Male', '2019-02-16', 'biraj.bhatta@email.com'),  
6 (100000005, 'Smarika Gurung', '2000-05-22', 'Female', '2020-02-16', 'smarika.gurung@email.com'),  
7 (100000006, 'Aaruhi KC', '1998-02-15', 'Female', '2018-02-16', 'aaruhi.kc@email.com'),  
8 (100000007, 'Pari GC', '2005-07-19', 'Female', '2022-09-16', 'pari.gc@email.com');
```

*Figure 24: Inserting into Students Table*

### Inserting into Program\_students Table

```
SQL> INSERT INTO Program_students (Program_id, Student_id)VALUES (1,1000000001);

1 row created.

SQL> INSERT INTO Program_students (Program_id, Student_id)VALUES (1,1000000002);
SQL> INSERT INTO Program_students (Program_id, Student_id)VALUES (2,1000000002);

1 row created.

SQL> INSERT INTO Program_students (Program_id, Student_id)VALUES (4,1000000003);

1 row created.

SQL> INSERT INTO Program_students (Program_id, Student_id)VALUES (5,1000000004);

1 row created.

SQL> INSERT INTO Program_students (Program_id, Student_id)VALUES (7,1000000005);

1 row created.

SQL> INSERT INTO Program_students (Program_id, Student_id)VALUES (6,1000000006);

1 row created.

SQL> INSERT INTO Program_students (Program_id, Student_id)VALUES (3,1000000007);

1 row created.

SQL> select * from Program_students;

PROGRAM_ID  STUDENT_ID
-----
1 1000000001
2 1000000002
3 1000000007
4 1000000003
5 1000000004
6 1000000006
```

Figure 25:Inserting into Program\_students Table

## Inserting into Modules Table

```
SQL> INSERT INTO Modules (Module_id, Module_name, Module_enrollment_capacity)
  2 VALUES ((4, 'Chemistry', 45);
VALUES ((4, 'Chemistry', 45)
      *
ERROR at line 2:
ORA-00907: missing right parenthesis

SQL> INSERT INTO Modules (Module_id, Module_name, Module_enrollment_capacity) VALUES (4, 'Chemistry', 45);
1 row created.

SQL> INSERT INTO Modules (Module_id, Module_name, Module_enrollment_capacity) VALUES (5, 'Biology', 55);
1 row created.

SQL> INSERT INTO Modules (Module_id, Module_name, Module_enrollment_capacity) VALUES(6, 'English Literature', 35);
1 row created.

SQL> INSERT INTO Modules (Module_id, Module_name, Module_enrollment_capacity) VALUES((7, 'History', 30);
INSERT INTO Modules (Module_id, Module_name, Module_enrollment_capacity) VALUES((7, 'History', 30)
      *
ERROR at line 1:
ORA-00907: missing right parenthesis

SQL> INSERT INTO Modules (Module_id, Module_name, Module_enrollment_capacity) VALUES (7, 'History', 30);
1 row created.

SQL> commit;
Commit complete.

SQL> |
```

Figure 26: Inserting into Modules Table

```
SQL> UPDATE MODULES
  2 SET Date_Announcements =
  3 CASE MODULE_ID
  4 WHEN 1 THEN TO_DATE('2024-05-01', 'YYYY-MM-DD')
  5 WHEN 2 THEN TO_DATE('2024-05-12', 'YYYY-MM-DD')
  6 WHEN 3 THEN TO_DATE('2024-05-15', 'YYYY-MM-DD')
  7 WHEN 4 THEN TO_DATE('2024-05-24', 'YYYY-MM-DD')
  8 WHEN 5 THEN TO_DATE('2024-05-26', 'YYYY-MM-DD')
  9 WHEN 6 THEN TO_DATE('2024-05-27', 'YYYY-MM-DD')
 10 WHEN 7 THEN TO_DATE('2024-05-28', 'YYYY-MM-DD')
 11 END;

7 rows updated.

SQL> |
```

Figure 27: Updating Module's Date\_Announcements row

```
SQL> UPDATE Modules
  2  SET Module_name = 'DATABASE'
  3  WHERE Module_name = 'IOT';

1 row updated.

SQL> |
```

Figure 28: Updating module's IOT row

### Inserting into Program\_Moduels Table

```
SQL> INSERT INTO Program_Modules (Program_id, Module_id) VALUES(1,1);
1 row created.

SQL> INSERT INTO Program_Modules (Program_id, Module_id) VALUES(2,2);
1 row created.

SQL> INSERT INTO Program_Modules (Program_id, Module_id) VALUES(3,3);
1 row created.

SQL> INSERT INTO Program_Modules (Program_id, Module_id) VALUES(4,1);
1 row created.

SQL> INSERT INTO Program_Modules (Program_id, Module_id) VALUES(5,6);
1 row created.

SQL>
SQL> SQL> INSERT INTO Program_Modules (Program_id, Module_id) VALUES(6,4);
SP2-0734: unknown command beginning "SQL> INSER..." - rest of line ignored.
SQL> INSERT INTO Program_Modules (Program_id, Module_id) VALUES(6,4);
1 row created.

SQL> INSERT INTO Program_Modules (Program_id, Module_id) VALUES(7,7);
1 row created.

SQL> |
```

Figure 29: Inserting into Program\_Moduels Table

## Inserting into Teachers Table

```
SQL> INSERT INTO Teachers (Teacher_id, Teacher_name, Teacher_email, Phone_number, Hire_date) VALUES (2, 'Dr. Sita Pandey', 'sita.pandey@university.com', '9807654321', TO_DATE('2016-02-14', 'YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Teachers (Teacher_id, Teacher_name, Teacher_email, Phone_number, Hire_date) VALUES (3, 'Dr. Krishna Gautam', 'krishna.gautam@university.com', '9812345678', TO_DATE('2017-07-23', 'YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Teachers (Teacher_id, Teacher_name, Teacher_email, Phone_number, Hire_date) VALUES (4, 'Dr. Anju Poudel', 'anju.poudel@university.com', '9804455667', TO_DATE('2016-11-05', 'YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Teachers (Teacher_id, Teacher_name, Teacher_email, Phone_number, Hire_date) VALUES (5, 'Dr. Kiran Thapa', 'kiran.thapa@university.com', '9885566778', TO_DATE('2019-09-30', 'YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Teachers (Teacher_id, Teacher_name, Teacher_email, Phone_number, Hire_date) VALUES (6, 'Dr. Sarita Koirala', 'sarita.koirala@university.com', '9806677889', TO_DATE('2020-01-10', 'YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Teachers (Teacher_id, Teacher_name, Teacher_email, Phone_number, Hire_date) VALUES (7, 'Dr. Binod Khadka', 'binod.khadka@university.com', '9807788990', TO_DATE('2014-04-20', 'YYYY-MM-DD'));
1 row created.

SQL>
```

Figure 30: Inserting into Teachers Table

## Inserting into Teacher\_Modules Table

```
SQL> INSERT INTO Teachers (Teacher_id, Teacher_name, Teacher_email, Phone_number, Hire_date) VALUES (7, 'Dr. Binod Khadka', 'binod.khadka@university.com', '9807788990', TO_DATE('2014-04-20', 'YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO Teacher_Modules (Module_id, Teacher_id)
2 VALUES(1, 1);
1 row created.

SQL> INSERT INTO Teacher_Modules (Module_id, Teacher_id) VALUES(2,2);
1 row created.

SQL> INSERT INTO Teacher_Modules (Module_id, Teacher_id) VALUES(3,3);
1 row created.

SQL> INSERT INTO Teacher_Modules (Module_id, Teacher_id) VALUES(4,4);
1 row created.

SQL> INSERT INTO Teacher_Modules (Module_id, Teacher_id) VALUES(5,5);
1 row created.

SQL> INSERT INTO Teacher_Modules (Module_id, Teacher_id) VALUES(6,6);
1 row created.

SQL> INSERT INTO Teacher_Modules (Module_id, Teacher_id) VALUES(7,7);
1 row created.

SQL>
```

Figure 31: Inserting into Teacher\_Modules Table

```
SQL> UPDATE TEACHER_MODULES
2 SET TEACHER_ID = 1
3 WHERE MODULE_ID = 4;

1 row updated.
```

Figure 32: Updating Teacher\_Modules table

```
SQL> UPDATE TEACHER_MODULES SET TEACHER_ID = 3 WHERE MODULE_ID = 5;

1 row updated.
```

Figure 33: Updating Teacher\_Modules II

## Inserting into Resources Table

```
SQL> INSERT INTO Resources (Resources_id, Resources_title, Resources_type, Resources_duration, Teacher_id, Module_id)VALUES(1, 'Mathematics Lecture Notes', 'Document', 2, 1, 1);

1 row created.

SQL> INSERT INTO Resources (Resources_id, Resources_title, Resources_type, Resources_duration, Teacher_id, Module_id)VALUES(2, 'Thermodynamics Lecture Slides', 'Presentation', 1, 2, 2);

1 row created.

SQL> INSERT INTO Resources (Resources_id, Resources_title, Resources_type, Resources_duration, Teacher_id, Module_id)VALUES (3, 'IOT Project Guide', 'Document', 3, 3, 3);

1 row created.

SQL> INSERT INTO Resources (Resources_id, Resources_title, Resources_type, Resources_duration, Teacher_id, Module_id)VALUES (4, 'Chemistry Experiment Video', 'Video', 4, 4, 4);

1 row created.

SQL> INSERT INTO Resources (Resources_id, Resources_title, Resources_type, Resources_duration, Teacher_id, Module_id)VALUES (5, 'Biology Lab Manual', 'Document', 5, 5, 5);

1 row created.

SQL> INSERT INTO Resources (Resources_id, Resources_title, Resources_type, Resources_duration, Teacher_id, Module_id)VALUES (6, 'English Literature Reading List', 'Document', 1, 6, 6);

1 row created.

SQL> INSERT INTO Resources (Resources_id, Resources_title, Resources_type, Resources_duration, Teacher_id, Module_id)VALUES (7, 'Electromagnetism Experiment Video', 'Video', 6, 7, 7);

1 row created.
```

Figure 34:Inserting into Resources Table

## Inserting into Teacher\_Resources Table

```
SQL> INSERT INTO Teacher_Resources (Teacher_id, Resources_id) VALUES (1, 1);

1 row created.

SQL> INSERT INTO Teacher_Resources (Teacher_id, Resources_id) VALUES (2, 2);

1 row created.

SQL> INSERT INTO Teacher_Resources (Teacher_id, Resources_id) VALUES (3,3);

1 row created.

SQL> INSERT INTO Teacher_Resources (Teacher_id, Resources_id) VALUES (4,4);

1 row created.

SQL> INSERT INTO Teacher_Resources (Teacher_id, Resources_id) VALUES (5,5);

1 row created.

SQL> INSERT INTO Teacher_Resources (Teacher_id, Resources_id) VALUES (6,6);

1 row created.

SQL> INSERT INTO Teacher_Resources (Teacher_id, Resources_id) VALUES (7,7);

1 row created.

SQL>
```

Figure 35:Inserting into Teacher\_Resources Table



## Inserting into Assessments Table

```
SQL> INSERT INTO Teacher_Resources (Teacher_id, Resources_id) VALUES (7,7);
1 row created.

SQL> INSERT INTO Assessments (Assessment_id, Assessment_title, Deadline, Weightage, Module_id)VALUES (1, 'Mathematics Midterm Exam', TO_DATE('2024-03-15', 'YYYY-MM-DD'), 40, 1);
1 row created.

SQL> INSERT INTO Assessments (Assessment_id, Assessment_title, Deadline, Weightage, Module_id)VALUES (2, 'Thermodynamics Quiz', TO_DATE('2024-04-10', 'YYYY-MM-DD'), 20, 2);
1 row created.

SQL> INSERT INTO Assessments (Assessment_id, Assessment_title, Deadline, Weightage, Module_id)VALUES (3, 'IOT Final Project', TO_DATE('2024-05-05', 'YYYY-MM-DD'), 50, 3);
1 row created.

SQL> INSERT INTO Assessments (Assessment_id, Assessment_title, Deadline, Weightage, Module_id)VALUES (4, 'Chemistry Lab Report', TO_DATE('2024-03-25', 'YYYY-MM-DD'), 30, 4);
1 row created.

SQL> INSERT INTO Assessments (Assessment_id, Assessment_title, Deadline, Weightage, Module_id)VALUES (5, 'Biology Field Study', TO_DATE('2024-04-20', 'YYYY-MM-DD'), 40, 5);
1 row created.

SQL> INSERT INTO Assessments (Assessment_id, Assessment_title, Deadline, Weightage, Module_id)VALUES (6, 'English Literature Essay', TO_DATE('2024-05-01', 'YYYY-MM-DD'), 25, 6);
1 row created.

SQL> INSERT INTO Assessments (Assessment_id, Assessment_title, Deadline, Weightage, Module_id)VALUES (7, 'Electromagnetism Lab Exam', TO_DATE('2024-03-30', 'YYYY-MM-DD'), 35, 7);
1 row created.

SQL>
```

*Figure 36:Inserting into Assessments Table*

## Inserting into Student\_assessments Table

```
SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES(2, 1000000004);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (3, 1000000005);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (3, 1000000006);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (4, 1000000007);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (4, 1000000001);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (5, 1000000002);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (5, 1000000003);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (6, 1000000004);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (6, 1000000005);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (7, 1000000006);
1 row created.

SQL> INSERT INTO Student_assessments (Assessment_id, Student_id) VALUES (7, 1000000007);
```

*Figure 37:Inserting into Student\_assessments Table*

## Inserting into Result Table

```
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000001, 1, 1, 85);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000001, 2, 1, 75);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000002, 1, 1, 90);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000002, 2, 1, 80);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000003, 3, 2, 70);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000003, 4, 2, 60);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000004, 3, 2, 50);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000004, 4, 2, 40);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000005, 5, 3, 30);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000005, 6, 3, 20);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000006, 5, 3, 45);
1 row created.
SQL> INSERT INTO RESULT (STUDENT_ID, ASSESSMENT_ID, PROGRAM_ID, GRADE) VALUES (1000000006, 6, 3, 55);
1 row created.
```

*Figure 38: Inserting into Result Table*

## Inserting into Announcements Table

```
SQL> INSERT INTO Announcements (Announcement_ID, Module_ID, Announcement_Date, Announcement_Text)
2  VALUES (1, 1, TO_DATE('2024-05-01', 'YYYY-MM-DD'), 'Midterm exam schedule released.');
```

1 row created.

```
SQL> INSERT INTO Announcements (Announcement_ID, Module_ID, Announcement_Date, Announcement_Text)
2  VALUES (2, 1, TO_DATE('2024-05-10', 'YYYY-MM-DD'), 'Project guidelines updated.')
```

3

```
SQL> INSERT INTO Announcements (Announcement_ID, Module_ID, Announcement_Date, Announcement_Text)
2  VALUES (2, 1, TO_DATE('2024-05-10', 'YYYY-MM-DD'), 'Project guidelines updated.');
```

1 row created.

```
SQL> INSERT INTO Announcements (Announcement_ID, Module_ID, Announcement_Date, Announcement_Text)
2  VALUES (3, 2, TO_DATE('2024-05-15', 'YYYY-MM-DD'), 'Guest lecture on Data Structures.');
```

1 row created.

```
SQL> INSERT INTO Announcements (Announcement_ID, Module_ID, Announcement_Date, Announcement_Text)
2  VALUES (4, 3, TO_DATE('2024-05-20', 'YYYY-MM-DD'), 'Final assessment details announced.');
```

1 row created.

Figure 39: Inserting Announcements table

## 7.Select Statement

### Displaying Programs Table

```
SQL> SELECT * FROM Programs;
```

| PROGRAM_ID | PROGRAM_NAME           | DEGREE_LEV |
|------------|------------------------|------------|
| 1          | Computer Science       | Bachelor   |
| 5          | Biotechnology          | Master     |
| 3          | Electrical Engineering | Master     |
| 2          | Mechanical Engineering | Bachelor   |
| 4          | Civil Engineering      | Bachelor   |
| 6          | Chemistry              | PhD        |
| 7          | Physics                | Bachelor   |

7 rows selected.

Figure 40: Displaying Programs Table

### Displaying Students Table

```
SQL> COLUMN STUDENT_DOB FORMAT A30;
SQL> SELECT * FROM Students;
```

| STUDENT_ID | STUDENT_NAME   | STUDENT_DOB | GENDER | STUDENT_ENROLLMENT_DATE | STUDENT_EMAIL            |
|------------|----------------|-------------|--------|-------------------------|--------------------------|
| 1000000001 | Aasal Paudel   | 15-MAY-00   | male   | 01-SEP-20               | aasal.paudel@email.com   |
| 1000000002 | Ramesh Khadka  | 22-NOV-03   | Male   | 05-SEP-20               | ramesh.khadka@email.com  |
| 1000000003 | Kritika Khadka | 02-SEP-04   | female | 10-SEP-21               | kritika.khadka@email.com |
| 1000000004 | Biraj Bhatta   | 06-MAR-00   | Male   | 16-FEB-19               | biraj.bhatta@email.com   |
| 1000000005 | Smarika Gurung | 22-MAY-00   | Female | 16-FEB-20               | smarika.gurung@email.com |
| 1000000006 | Aaruhi KC      | 15-FEB-98   | Female | 16-FEB-18               | aaruhi.kc@email.com      |
| 1000000007 | Pari GC        | 19-JUL-05   | Female | 16-SEP-22               | pari.gc@email.com        |

7 rows selected.

SQL>

Figure 41: Displaying Students Table

**Displaying Program\_students Table**

```
SQL> select * from Program_students;
```

| PROGRAM_ID | STUDENT_ID |
|------------|------------|
| 1          | 1000000001 |
| 2          | 1000000002 |
| 3          | 1000000007 |
| 4          | 1000000003 |
| 5          | 1000000004 |
| 6          | 1000000006 |
| 7          | 1000000005 |

```
7 rows selected.
```

```
SQL> |
```

Figure 42:Displaying Program\_students Table

**Displaying Modules Table**

```
SQL> SELECT * FROM MODULES;
```

| MODULE_ID | MODULE_NAME        | MODULE_ENROLLMENT_CAPACITY | DATE_ANNO |
|-----------|--------------------|----------------------------|-----------|
| 1         | Mathematics        | 50                         | 01-MAY-24 |
| 2         | Thermodynamics     | 40                         | 12-MAY-24 |
| 3         | DATABASE           | 40                         | 15-MAY-24 |
| 4         | Chemistry          | 45                         | 24-MAY-24 |
| 5         | Biology            | 55                         | 26-MAY-24 |
| 6         | English Literature | 35                         | 27-MAY-24 |
| 7         | Electromagnetism   | 30                         | 28-MAY-24 |

```
7 rows selected.
```

```
SQL> |
```

Figure 43:Displaying Modules Table

**Displaying Program\_Moduels Table**

```
SQL> select * from PROGRAM_MODULES;
```

| PROGRAM_ID | MODULE_ID |
|------------|-----------|
| 1          | 1         |
| 2          | 2         |
| 3          | 3         |
| 4          | 1         |
| 5          | 6         |
| 6          | 4         |
| 7          | 7         |

```
7 rows selected.

SQL> |
```

*Figure 44:Displaying Program\_Moduels Table***Displaying Teachers Table**

```
SQL> COLUMN TEACHER_EMAIL FORMAT A30;
SQL> select * from TEACHERS;
```

| TEACHER_ID | TEACHER_NAME       | TEACHER_EMAIL                 | PHONE_NUMB | HIRE_DATE |
|------------|--------------------|-------------------------------|------------|-----------|
| 1          | Dr. Suman Acharya  | suman.acharya@university.com  | 9801234567 | 12-AUG-15 |
| 2          | Dr. Sita Pandey    | sita.pandey@university.com    | 9807654321 | 14-FEB-16 |
| 3          | Dr. Krishna Gautam | krishna.gautam@university.com | 9812345678 | 23-JUL-17 |
| 4          | Dr. Anju Poudel    | anju.poudel@university.com    | 9804455667 | 05-NOV-16 |
| 5          | Dr. Kiran Thapa    | kiran.thapa@university.com    | 9805566778 | 30-SEP-19 |
| 6          | Dr. Sarita Koirala | sarita.koirala@university.com | 9806677889 | 10-JAN-20 |
| 7          | Dr. Binod Khadka   | binod.khadka@university.com   | 9807788990 | 20-APR-14 |

```
7 rows selected.

SQL>
```

*Figure 45:Displaying Teachers Table*

### Displaying Teacher\_Modules Table

```
SQL> SELECT * FROM TEACHER_MODULES;
```

| MODULE_ID | TEACHER_ID |
|-----------|------------|
| 1         | 1          |
| 2         | 2          |
| 3         | 3          |
| 4         | 1          |
| 5         | 3          |
| 6         | 6          |
| 7         | 7          |

7 rows selected.

Figure 46: Displaying Teacher\_Modules Table

### Displaying Resources Table

```
SQL> COLUMN RESOURCES_TITLE FORMAT A35;
SQL> select * from RESOURCES;
```

| RESOURCES_ID | RESOURCES_TITLE                   | RESOURCES_TY | RESOURCES_DURATION | TEACHER_ID | MODULE_ID |
|--------------|-----------------------------------|--------------|--------------------|------------|-----------|
| 1            | Mathematics Lecture Notes         | Document     | 2                  | 1          | 1         |
| 2            | Thermodynamics Lecture Slides     | Presentation | 1                  | 2          | 2         |
| 3            | IOT Project Guide                 | Document     | 3                  | 3          | 3         |
| 4            | Chemistry Experiment Video        | Video        | 4                  | 4          | 4         |
| 5            | Biology Lab Manual                | Document     | 5                  | 5          | 5         |
| 6            | English Literature Reading List   | Document     | 1                  | 6          | 6         |
| 7            | Electromagnetism Experiment Video | Video        | 6                  | 7          | 7         |

7 rows selected.

```
SQL> |
```

Figure 47: Displaying Resources Table

**Displaying Teacher\_Resources Table**

```
SQL> select * from TEACHER_RESOURCES;
```

| TEACHER_ID | RESOURCES_ID |
|------------|--------------|
| 1          | 1            |
| 2          | 2            |
| 3          | 3            |
| 4          | 4            |
| 5          | 5            |
| 6          | 6            |
| 7          | 7            |

```
7 rows selected.

SQL> |
```

*Figure 48:Displaying Teacher\_Resources Table***Displaying Assessments Table**

```
SQL> COLUMN ASSESSMENT_TITLE FORMAT A30;
SQL> select * from ASSESSMENTS;
```

| ASSESSMENT_ID | ASSESSMENT_TITLE          | DEADLINE  | WEIGHTAGE | MODULE_ID |
|---------------|---------------------------|-----------|-----------|-----------|
| 1             | Mathematics Midterm Exam  | 15-MAR-24 | 40        | 1         |
| 2             | Thermodynamics Quiz       | 10-APR-24 | 20        | 2         |
| 3             | IOT Final Project         | 05-MAY-24 | 50        | 3         |
| 4             | Chemistry Lab Report      | 25-MAR-24 | 30        | 4         |
| 5             | Biology Field Study       | 20-APR-24 | 40        | 5         |
| 6             | English Literature Essay  | 01-MAY-24 | 25        | 6         |
| 7             | Electromagnetism Lab Exam | 30-MAR-24 | 35        | 7         |

```
7 rows selected.

SQL>
```

*Figure 49:Displaying Assessments Table*



**Displaying Student\_assessments Table**

```
SQL> select * from STUDENT_ASSESSMENTS;
```

| STUDENT_ID | ASSESSMENT_ID |
|------------|---------------|
| -----      | -----         |
| 1000000001 | 1             |
| 1000000001 | 4             |
| 1000000002 | 1             |
| 1000000002 | 5             |
| 1000000003 | 2             |
| 1000000003 | 5             |
| 1000000004 | 2             |
| 1000000004 | 6             |
| 1000000005 | 3             |
| 1000000005 | 6             |
| 1000000006 | 3             |
| 1000000006 | 7             |
| 1000000007 | 4             |
| 1000000007 | 7             |

```
14 rows selected.
```

```
SQL> |
```

*Figure 50: Displaying Students\_assessments Table*

## Displaying Result Table

```
SQL> SELECT * FROM RESULT;
```

| STUDENT_ID | ASSESSMENT_ID | PROGRAM_ID | GRADE | MODULE_ID |
|------------|---------------|------------|-------|-----------|
| 1000000001 | 1             | 1          | 85    | 1         |
| 1000000001 | 2             | 1          | 75    | 1         |
| 1000000002 | 1             | 1          | 90    | 1         |
| 1000000002 | 2             | 1          | 80    | 1         |
| 1000000003 | 3             | 2          | 70    | 2         |
| 1000000003 | 4             | 2          | 60    | 2         |
| 1000000004 | 3             | 2          | 50    | 2         |
| 1000000004 | 4             | 2          | 40    | 2         |
| 1000000005 | 5             | 3          | 30    | 3         |
| 1000000005 | 6             | 3          | 20    | 3         |
| 1000000006 | 5             | 3          | 45    | 3         |

| STUDENT_ID | ASSESSMENT_ID | PROGRAM_ID | GRADE | MODULE_ID |
|------------|---------------|------------|-------|-----------|
| 1000000006 | 6             | 3          | 55    | 3         |

12 rows selected.

```
SQL> |
```

Figure 51: Displaying Result Table

## Displaying Announcements Table

```
SQL> COLUMN ANNOUNCEMENT_TEXT FORMAT A27;
```

```
SQL> SELECT * FROM ANNOUNCEMENTS;
```

| ANNOUNCEMENT_ID | MODULE_ID | ANNOUNCEMENT_DATE | ANNOUNCEMENT_TEXT                   |
|-----------------|-----------|-------------------|-------------------------------------|
| 1               | 1         | 01-MAY-24         | Midterm exam schedule released.     |
| 2               | 1         | 10-MAY-24         | Project guidelines updated.         |
| 3               | 2         | 15-MAY-24         | Guest lecture on Data Structures.   |
| 4               | 3         | 20-MAY-24         | Final assessment details announced. |

```
SQL> |
```

Figure 52: Displaying Announcements table

## 8.Information and Transaction Query

### 8.1Information query

1. List the programs that are available in the college and the total number of students enrolled in each.

```
SQL> SELECT
  2     p.PROGRAM_NAME,
  3     COUNT(ps.STUDENT_ID) AS Total_Students_Enrolled
  4 FROM
  5     PROGRAMS p
  6 JOIN
  7     PROGRAM_STUDENTS ps ON p.PROGRAM_ID = ps.PROGRAM_ID
  8 GROUP BY
  9     p.PROGRAM_NAME
 10 ORDER BY
 11     p.PROGRAM_NAME;
```

| PROGRAM_NAME           | TOTAL_STUDENTS_ENROLLED |
|------------------------|-------------------------|
| Biotechnology          | 1                       |
| Chemistry              | 1                       |
| Civil Engineering      | 1                       |
| Computer Science       | 1                       |
| Electrical Engineering | 1                       |
| Mechanical Engineering | 1                       |
| Physics                | 1                       |

7 rows selected.

```
SQL> |
```

Figure 53: Information query1

2. List all the announcements made for a particular module starting from 1<sup>st</sup> May 2024 to 28<sup>th</sup> May 2024.

```
SQL> COLUMN ANNOUNCEMENT_TEXT FORMAT A31;
SQL> SELECT Announcement_ID, Module_ID, Announcement_Date, Announcement_Text
 2  FROM Announcements
 3  WHERE Module_ID = 1
 4    AND Announcement_Date BETWEEN TO_DATE('2024-05-01', 'YYYY-MM-DD')
 5                                AND TO_DATE('2024-05-28', 'YYYY-MM-DD')
 6  ORDER BY Announcement_Date;
```

| ANNOUNCEMENT_ID | MODULE_ID | ANNOUNCEM | ANNOUNCEMENT_TEXT               |
|-----------------|-----------|-----------|---------------------------------|
| 1               | 1         | 01-MAY-24 | Midterm exam schedule released. |
| 2               | 1         | 10-MAY-24 | Project guidelines updated.     |

```
SQL> |
```

Figure 54: Information query 2

3. List the names of all modules that begin with the letter 'D', along with the total number of resources uploaded for those modules.

```
SQL> SELECT
2      M.MODULE_NAME,
3      COUNT(R.RESOURCES_ID) AS TOTAL_RESOURCES
4  FROM
5      MODULES M
6  LEFT JOIN
7      RESOURCES R
8  ON
9      M.MODULE_ID = R.MODULE_ID
10 WHERE
11     M.MODULE_NAME LIKE 'D%'
12 GROUP BY
13     M.MODULE_NAME
14 ORDER BY
15     M.MODULE_NAME;
```

| MODULE_NAME | TOTAL_RESOURCES |
|-------------|-----------------|
| -----       | -----           |
| DATABASE    | 1               |

SQL> |

Figure 55: Information query 3

4. List the names of all students along with their enrolled who have not submitted any assessments for a particular module.

```
SQL> SELECT S.STUDENT_NAME, P.PROGRAM_NAME, M.MODULE_NAME FROM STUDENTS S JOIN PROGRAM_STUDENTS PS ON S.STUDENT_ID = PS.STUDENT_ID JOIN PROGRAMS P ON PS.PROGRAM_ID = P.PROGRAM_ID JOIN PROGRAM_MODULES PM ON P.PROGRAM_ID = PM.PROGRAM_ID JOIN MODULES M ON PM.MODULE_ID = M.MODULE_ID WHERE M.MODULE_ID = 1 AND S.STUDENT_ID NOT IN (SELECT R.STUDENT_ID FROM RESULT R WHERE R.MODULE_ID = M.MODULE_ID) ORDER BY S.STUDENT_NAME;
```

| STUDENT_NAME   | PROGRAM_NAME      | MODULE_NAME |
|----------------|-------------------|-------------|
| -----          | -----             | -----       |
| Kritika Khadka | Civil Engineering | Mathematics |

SQL> |

Figure 56: Information query 4

5. List all the teachers who teach more than one module.

| TEACHER_ID | MODULECOUNT |
|------------|-------------|
| 1          | 2           |
| 3          | 2           |

```
SQL> |
```

Figure 57: Information query 5

## 8.2. Transaction query

1. Identify the module that has the latest assessment deadline.

```
SQL> SELECT MODULE_ID
2 FROM ASSESSMENTS
3 WHERE DEADLINE = (SELECT MAX(DEADLINE) FROM ASSESSMENTS);

MODULE_ID
-----
3

SQL> |
```

Figure 58: Transaction query 1

- Find the top three students who have the highest total score across all modules.

```
SQL> SELECT * FROM (SELECT r.STUDENT_ID,SUM(r.GRADE) AS total_score FROM RESULT r GROUP BY r.STUDENT_ID ORDER BY total_score DESC) WHERE ROWNUM <= 3;
```

| STUDENT_ID | TOTAL_SCORE |
|------------|-------------|
| 1000000002 | 170         |
| 1000000001 | 160         |
| 1000000003 | 130         |

```
SQL> |
```

Figure 59: Transaction query 2

- Find the total number of assignments for each program and the average score across all assessments in those programs.

```
SQL> SELECT
2      r.PROGRAM_ID,
3      COUNT(r.ASSESSMENT_ID) AS total_assessments,
4      AVG(r.GRADE) AS average_score
5  FROM
6      RESULT r
7  GROUP BY
8      r.PROGRAM_ID
9  ORDER BY
10     r.PROGRAM_ID;
```

| PROGRAM_ID | TOTAL_ASSESSMENTS | AVERAGE_SCORE |
|------------|-------------------|---------------|
| 1          | 4                 | 82.5          |
| 2          | 4                 | 55            |
| 3          | 4                 | 37.5          |

```
SQL> |
```

Figure 60: Transaction query 3

4. List the students who have scored above the average score in 'Databases' module.

```
SQL> SELECT
  2     r.STUDENT_ID,
  3     AVG(r.GRADE) AS Average_Score
  4 FROM
  5     RESULT r
  6 WHERE
  7     r.MODULE_ID = 3
  8 GROUP BY
  9     r.STUDENT_ID
 10 HAVING
 11     AVG(r.GRADE) > (SELECT AVG(GRADE) FROM RESULT WHERE MODULE_ID = 3);

STUDENT_ID AVERAGE_SCORE
-----
1000000006          50

SQL>
```

Figure 61: Transaction query 4



5. Display whether a student has passed or failed as remarks as per their total aggregate marks obtained in a particular module. (NOTE: Consider total aggregate marks equal to or above 40 is pass, below 40 is fail)

```
SQL> SELECT
  2     r.STUDENT_ID,
  3     r.MODULE_ID,
  4     SUM(r.GRADE) AS Total_Aggregate_Marks,
  5     CASE
  6         WHEN SUM(r.GRADE) >= 40 THEN 'Pass'
  7         ELSE 'Fail'
  8     END AS Remarks
  9 FROM
 10     RESULT r
 11 GROUP BY
 12     r.STUDENT_ID, r.MODULE_ID;
```

| STUDENT_ID | MODULE_ID | TOTAL_AGGREGATE_MARKS | REMA |
|------------|-----------|-----------------------|------|
| 1000000001 | 1         | 160                   | Pass |
| 1000000002 | 1         | 170                   | Pass |
| 1000000006 | 3         | 100                   | Pass |
| 1000000004 | 2         | 90                    | Pass |
| 1000000003 | 2         | 130                   | Pass |
| 1000000005 | 3         | 50                    | Pass |

6 rows selected.

```
SQL> |
```

Figure 62: Transaction query5

## 9.Creating Dump file and Drop Table

### Dump File:

```
C:\Users\Tilasm Subedi> Exp Eclassroom/tilasmi file = coursework.dmp

Export: Release 11.2.0.2.0 - Production on Wed Jan 22 18:28:11 2025

Copyright (c) 1982, 2009, Oracle and/or its affiliates. All rights reserved.

Connected to: Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production
Export done in WE8MSWIN1252 character set and AL16UTF16 NCHAR character set
server uses AL32UTF8 character set (possible charset conversion)
. exporting pre-schema procedural objects and actions
. exporting foreign function library names for user ECLASSROOM
. exporting PUBLIC type synonyms
. exporting private type synonyms
. exporting object type definitions for user ECLASSROOM
About to export ECLASSROOM's objects ...
. exporting database links
. exporting sequence numbers
. exporting cluster definitions
. about to export ECLASSROOM's tables via Conventional Path ...
. . exporting table ANNOUNCEMENTS 4 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table ASSESSMENTS 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table MODULES 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table PROGRAMS 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table PROGRAM_MODULES 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table PROGRAM_STUDENTS 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table RESOURCES 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table RESULT 12 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table STUDENTS 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table STUDENT_ASSESSMENTS 14 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table TEACHERS 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table TEACHER_MODULES 7 rows exported
EXP-00091: Exporting questionable statistics.
. . exporting table TEACHER_RESOURCES 7 rows exported
EXP-00091: Exporting questionable statistics.
. exporting synonyms
. exporting views
. exporting stored procedures
. exporting operators
. exporting referential integrity constraints
. exporting triggers
. exporting indextypes
. exporting bitmap, functional and extensible indexes
. exporting posttables actions
. exporting materialized views
. exporting snapshot logs
. exporting job queues
. exporting refresh groups and children
. exporting dimensions
. exporting post-schema procedural objects and actions
. exporting statistics
Export terminated successfully with warnings.

C:\Users\Tilasm Subedi>
```

Figure 63: Dump file creation

**DROP TABLE:**

```
SQL> DROP TABLE PROGRAM_STUDENTS;
Table dropped.

SQL> DROP TABLE PROGRAM_MODULES;
Table dropped.

SQL> DROP TABLE TEACHER_MODULES;
Table dropped.

SQL> DROP TABLE TEACHER_RESOURCES;
Table dropped.

SQL> DROP TABLE STUDENT_ASSESSMENTS;
Table dropped.

SQL> DROP TABLE PROGRAMS;
DROP TABLE PROGRAMS
*
ERROR at line 1:
ORA-02449: unique/primary keys in table referenced by foreign keys

SQL> DROP TABLE ANNOUNCEMENTS;
Table dropped.

SQL> DROP TABLE RESULT;
Table dropped.

SQL> DROP TABLE ASSESSMENTS;
Table dropped.

SQL> DROP TABLE RESOURCES;
Table dropped.

SQL> DROP TABLE TEACHERS;
Table dropped.

SQL> DROP TABLE MODULES;
Table dropped.

SQL> DROP TABLE PROGRAMS;
DROP TABLE PROGRAMS
*
ERROR at line 1:
ORA-02449: unique/primary keys in table referenced by foreign keys

SQL> DROP TABLE STUDENTS;
Table dropped.

SQL> DROP TABLE PROGRAMS;
Table dropped.

SQL> |
```

*Figure 64: Dropping all the tables*

## 10.Data Dictionary

### Programs Table

*Table 3:Programs data dictionary*

| Attribute         | Data Type | Description                                       | Length | Key     | Null | Unique |
|-------------------|-----------|---------------------------------------------------|--------|---------|------|--------|
| <u>Program_id</u> | INT       | Unique identifier for each program.               | 10     | Primary | NO   | YES    |
| Program_name      | VARCHAR   | Name of program.                                  | 25     |         | NO   | NO     |
| Degree_level      | VARCHAR   | Level of the degree (e.g., Bachelor's, Master's). | 25     |         | NO   | NO     |

### Students Table

*Table 4:Students data dictionary*

| Attribute         | Data Type | Description                         | Length | Key     | Null |
|-------------------|-----------|-------------------------------------|--------|---------|------|
| <u>Student_id</u> | INT       | Unique identifier for each student. | 10     | Primary | NO   |
| Student_name      | VARCHAR   | Full name of the student.           | 25     |         | NO   |
| Student_DOB       | DATE      | Date of the birth of the student.   |        |         | NO   |

|                         |         |                               |    |  |    |
|-------------------------|---------|-------------------------------|----|--|----|
| Gender                  | VARCHAR | Gender of the student         | 10 |  | NO |
| Student_Enrollment_Date | DATE    | Date the student enrolled.    |    |  | NO |
| Student_email           | VARCHAR | Email address of the student. | 30 |  | NO |

### Program\_students Table

*Table 5:Program\_students data dictionary*

| Field Name        | Data Type | Description                                 | Length | Key     | Null | Unique |
|-------------------|-----------|---------------------------------------------|--------|---------|------|--------|
| <u>Program_id</u> | INT       | Composite primary key referencing Programs. | 10     | Foreign | NO   | NO     |
| <u>Student_id</u> | INT       | Composite primary referencing Students.     | 10     | Foreign | NO   | NO     |

**Modules Table***Table 6: Modules data dictionary*

| Attribute                  | Data Type | Description                            | Length | Key     | Null | Unique |
|----------------------------|-----------|----------------------------------------|--------|---------|------|--------|
| <u>Module_id</u>           | INT       | Unique identifier for each module.     | 10     | Primary | NO   | YES    |
| Module_name                | VARCHAR   | Name of the module.                    | 20     |         | NO   | NO     |
| Module_Enrollment_Capacity | INT       | Maximum number of students per module. | 26     |         | NO   | NO     |
| Date_Announcement          | DATE      | Date of announcement.                  | 10     |         | NO   | NO     |

**Program\_Modules Table***Table 7: Program\_Modules data dictionary*

| Attribute         | Data Type | Description                                 | Length | Key     | Null | Unique |
|-------------------|-----------|---------------------------------------------|--------|---------|------|--------|
| <u>Program_id</u> | INT       | Composite primary key referencing Programs. | 10     | Foreign | NO   | NO     |
| <u>Module_id</u>  | INT       | Composite primary key referencing Modules.  | 10     | Foreign | NO   | NO     |

**Teacher Table***Table 8: Teacher data dictionary*

| Attribute         | Data Type | Description                         | Length | Key     | Null | Unique |
|-------------------|-----------|-------------------------------------|--------|---------|------|--------|
| <u>Teacher_id</u> | INT       | Unique identifier for each teacher. | 10     | Primary | NO   | YES    |
| Teacher_name      | VARCHAR   | Full name of the teacher.           | 25     |         | NO   | NO     |
| Teacher_email     | VARCHAR   | Email address of the teacher.       | 30     |         | NO   | YES    |
| Phone_number      | VARCHAR   | Phone number of the teacher.        | 15     |         | NO   | NO     |
| Hire_date         | DATE      | Date the teacher was hired.         |        |         | NO   | NO     |

**Teacher\_Modules Table***Table 9:Teacher\_Modules data dictionary*

| Attribute         | Data Type | Description                                 | Length | Key     | Null | Unique |
|-------------------|-----------|---------------------------------------------|--------|---------|------|--------|
| <u>Module_id</u>  | INT       | Composite primary key referencing Modules.  | 10     | Foreign | NO   | YES    |
| <u>Teacher_id</u> | INT       | Composite primary key referencing Teachers. | 10     | Foreign | NO   | YES    |



**Resources Table***Table 10:Resources data dictionary*

| Field Name          | Data Type | Description                           | Length | Key     | Null | Unique |
|---------------------|-----------|---------------------------------------|--------|---------|------|--------|
| <u>Resources_id</u> | INT       | Unique identifier for each resource.  | 10     | Primary | NO   | YES    |
| Resources_title     | VARCHAR   | Title of the resource.                | 25     |         | NO   | NO     |
| Resources_type      | VARCHAR   | Type of resource (e.g., video. Book). | 12     |         | NO   | NO     |
| Resources_duration  | INT       | Duration of the resource              | 10     |         | NO   | NO     |
| Teacher_id          | INT       | Foreign key referencing Teachers.     |        | Foreign | NO   | YES    |
| Module_id           | INT       | Foreign key referencing modules.      |        | Foreign | NO   | YES    |

**Teacher\_Resources Table***Table 11:Teacher\_Resources data dictionary*

| Field Name          | Data Type | Description                                  | Length | Key     | Null | Unique |
|---------------------|-----------|----------------------------------------------|--------|---------|------|--------|
| <u>Teacher_id</u>   | INT       | Composite primary key referencing Teachers.  | 10     | Foreign | NO   | YES    |
| <u>Resources_id</u> | INT       | Composite primary key referencing Resources. | 10     | Foreign | NO   | YES    |

**Assessments Table***Table 12:Assessments data dictionary*

| Attribute            | Data Type | Description                            | Length | Key     | Null | Unique |
|----------------------|-----------|----------------------------------------|--------|---------|------|--------|
| <u>Assessment_id</u> | INT       | Unique identifier for each assessment. | 10     | Primary | NO   | YES    |
| Assessment_title     | VARCHAR   | Title of the assessment.               | 10     |         | NO   | NO     |
| Deadline             | DATE      | Due date for the assessment.           |        |         | NO   | NO     |
| Weightage            | DECIMAL   | Percentage weight of the assessment.   | 10     |         | NO   | NO     |

|                  |     |                                        |    |         |    |     |
|------------------|-----|----------------------------------------|----|---------|----|-----|
| <u>Module_id</u> | INT | Foreign key<br>referencing<br>Modules. | 10 | Foreign | NO | YES |
|------------------|-----|----------------------------------------|----|---------|----|-----|

### Students\_Assessments Table

*Table 13: Students\_Assessments data dictionary*

| Attribute            | Data Type | Description                                          | Length | Key     | Null | Unique |
|----------------------|-----------|------------------------------------------------------|--------|---------|------|--------|
| <u>Assessment_id</u> | INT       | Composite primary key<br>referencing<br>Assessments. | 10     | Foreign | NO   | YES    |
| <u>Student_id</u>    | INT       | Composite primary key<br>referencing<br>Students.    | 10     | Foreign | NO   | YES    |

**Result Table***Table 14:Result data dictionary*

| Attribute            | Data Type | Description                          | Length | Key     | Null | Unique |
|----------------------|-----------|--------------------------------------|--------|---------|------|--------|
| <u>Student_id</u>    | INT       | Foreign key referencing Students.    | 10     | Foreign | NO   | YES    |
| <u>Assessment_id</u> | INT       | Foreign key referencing Assessments. | 10     | Foreign | NO   | YES    |
| <u>Program_id</u>    | INT       | Foreign key referencing Programs.    | 10     | Foreign | NO   | YES    |
| Grade                | INT       | Grade obtained by the student.       | 10     |         | NO   | NO     |
| Module_id            | INT       | Foreign key referencing Modules.     | 10     | Foreign | NO   | YES    |

**Announcements Table***Table 15:Announcements table*

| Attribute              | Data Type | Description                              | Length | Key     | Null | Unique |
|------------------------|-----------|------------------------------------------|--------|---------|------|--------|
| <u>Announcement_id</u> | INT       | Unique identifier for each announcement. | 10     | Primary | NO   | YES    |
| Module_id              | INT       | Foreign key referencing Modules.         | 10     | Foreign | NO   | YES    |
| Announcement_text      | VARCHAR   | Date on which the announcement was made. | 27     |         | NO   | NO     |
| Date_Announcement      | DATE      | Date of announcement.                    | 10     |         | NO   | NO     |

## 9. Conclusion

This coursework has been an insightful and enriching journey, offering a hands-on opportunity to apply database design principles to a practical scenario. The project revolves around creating a robust database system for the “E-Classroom” platform, designed to streamline the management of students, teachers, programs, modules, and assessments. By working on this case study, a deeper understanding of the real-world applications of database systems has been achieved.

The process included designing and refining Entity Relationship Diagrams (ERDs), normalizing data into Third Normal Form (3NF), and writing efficient SQL queries to support the platform’s functionality. Challenges, such as maintaining data integrity and ensuring efficient query execution, were navigated successfully, fostering problem-solving skill and technical expertise.

The case study highlighted the importance of structured data management in educational platforms, enabling features like tracking student progress, managing module resources, and ensuring a logical sequence of resource completion. This practical context underscored the significance of database systems in creating a dynamic and efficient digital environment.

Beyond fulfilling the coursework objectives, this experience has been instrumental in strengthening technical and analytical skills while demonstrating the broader impact of database systems in addressing complex organizational needs. The knowledge and expertise gained from this project provide a strong foundation for future academic pursuits and professional endeavours in database design and management.

## 9. References

- Garge, S., 2024. *Alchetron*. [Online]  
Available at: <https://alchetron.com/Unnormalized-form>  
[Accessed 09 October 2024].
- h. et al., 2024. *Microsoft Learn Challenge*. [Online]  
Available at: <https://learn.microsoft.com/en-us/office/troubleshoot/access/database-normalization-description>  
[Accessed 23 November 2024].
- Melanie, 2024. *DataScientest*. [Online]  
Available at: <https://datascientest.com/en/primary-key-definition-advantages-and-special-features>  
[Accessed 23 April 2024].
- Oyairo, A., 2024. *builtin*. [Online]  
Available at: <https://builtin.com/articles/entity-relationship-diagram>  
[Accessed 3 June 2024].
- S, R. A., 2024. *simplilearn*. [Online]  
Available at: <https://www.simplilearn.com/tutorials/sql-tutorial/er-diagram-in-dbms>  
[Accessed 2 December 2024].